CEITEC

MUNI



C9555: A survival guide to simulations

Week 2

Denys Biriukov – Oct. 5th, 2023

Get ready for practical session

- Open new terminal `Ctrl + Alt + T`
- Course materials at: https://gitlab.com/denysbiriukov/survival_guide_simulations_c9555
- MetaCentrum registration at: https://metavo.metacentrum.cz/en/application/index.html
- Any technical issues/problems/questions? Checklist:
  - Access to Wolf
  - MetaCentrum account
  - Proper infinity and conda environments (we will adjust it before practical session)

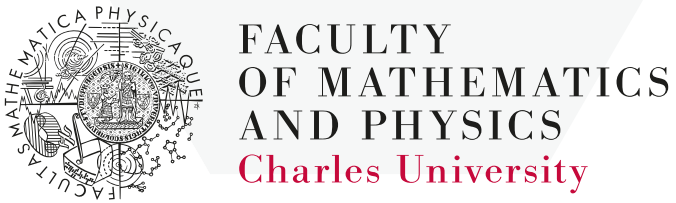
Acknowledgments

This part of the course is inspired and partially based on:

- Teaching materials of “Spring School in Computational Chemistry 2023” taken place in Finland and prepared by Dr. Hector Martinez-Seara and Dr. Matti Javanainen.
- Teaching course about molecular dynamics simulations at Charles University by prof. Lukasz Cwiklik.







More a discussion than monologue

If you have a quick question to catch up, raise your hand and ask – we will try to fix it and bring you back

If quick answer does not help, let's chat during a break or after

This course is heavily practical – theory will be trying to catch up, not vice versa

What is today?

- Week 2:
  Mixing Solvent – Physics and Chemistry in a Box

- Any simulation course has a fundamental (basic, starting-point) part.
- You may also call it a “boring” part. ☺

- “We must learn to walk before we can run”.

TEACHING WEEK	DATE	LECTURER	TOPIC
01	21.09	TR	Getting Started (Linux, Python, VMD)
02	05.10	DB	Mixing Solvents – Physics and Chemistry in a Box – part I
03	12.10	DB	Mixing Solvents – Physics and Chemistry in a Box – part II
04	19.10	DB	Ions, Molecules, and Their Interactions
05	26.10	OK	Making Drugs *in silico* – part I – Lipid Membranes
06	02.11	OK	Making Drugs *in silico* – part II – Ligands Parametrizations
07	09.11	TR	Making Drugs *in silico* – part III – Steered Molecular Dynamics & Free Energy Calculations
08	16.11	TR	Making Drugs *in silico* – part IV – Free Energy Calculations & Permeability Coefficient
09	23.11	WM	How to Beat AlphaFold – part I
10	30.11	WM	How to Beat AlphaFold – part II
11	07.12	WM	Working in a Professional Environment
12	14.12	OK	Zápočet – first session to start your project

Today program

Lecture (1st hour)

- Basics of molecular dynamics simulations
- Why others need it and what physics is behind
- Time in simulations – concepts and must-know

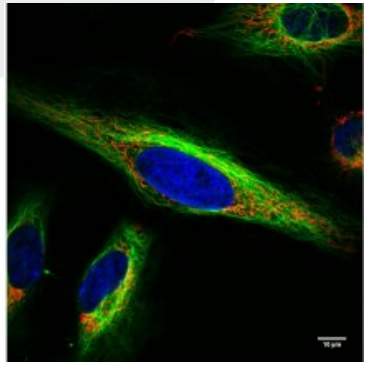
Practical session (2nd and 3rd hours)

- GROMACS – what is it and how to use it?
- Basic tools to build a system and run a simulation in GROMACS
- Preparing example system put in the context of the lecture

Synergy of experiments, simulations, and theory

Physical system
(cluster of atoms, water-air interface,
cell membrane, protein in water)

Experiment
(IR spectroscopy, NMR,
X-ray diffraction,
fluorescence microscopy,
enzymatic kinetics)



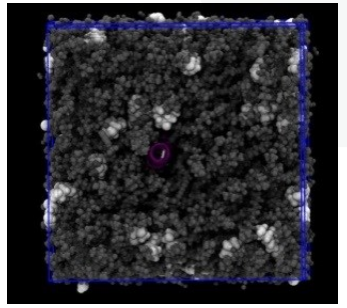
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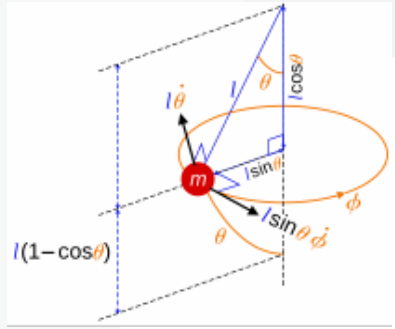
Theory
(quantum mechanics,
thermodynamics,
chemical kinetics)



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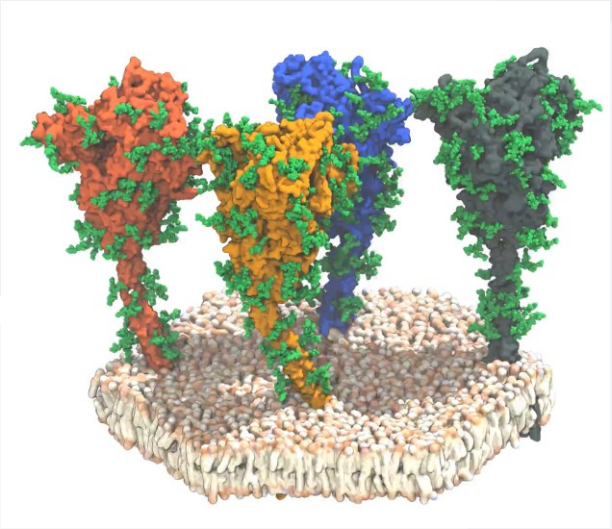
Simulations
(molecular dynamics, Monte Carlo, dissipative particle dynamics, computational fluid dynamics)





Molecular Simulations: Physics, not Cartoon

- Why do we actually use simulations?
  - Visualizing experiments? Reproducing? Confirming?
- A molecular simulation is a well-controlled *in silico* experiment itself!
  - “Computational microscope”, “computational experiment”
- Good simulation model – predictive power of simulations
  - Simulations can initiate new experimental work
  - Certain computational screenings of novel drugs are more efficient (and much cheaper) than many (if not all) experiments
- Good model = fundamental physics behind it (approximation of fundamental physical laws)



Spike protein of SARS-CoV-2

TUROŇOVÁ, SIKORA, SCHÜRMANN et al.
MAX PLANCK INSTITUTE OF BIOPHYSICS
PAUL-EHRLICH-INSTITUT

Molecular Simulations: Physics, not Cartoon

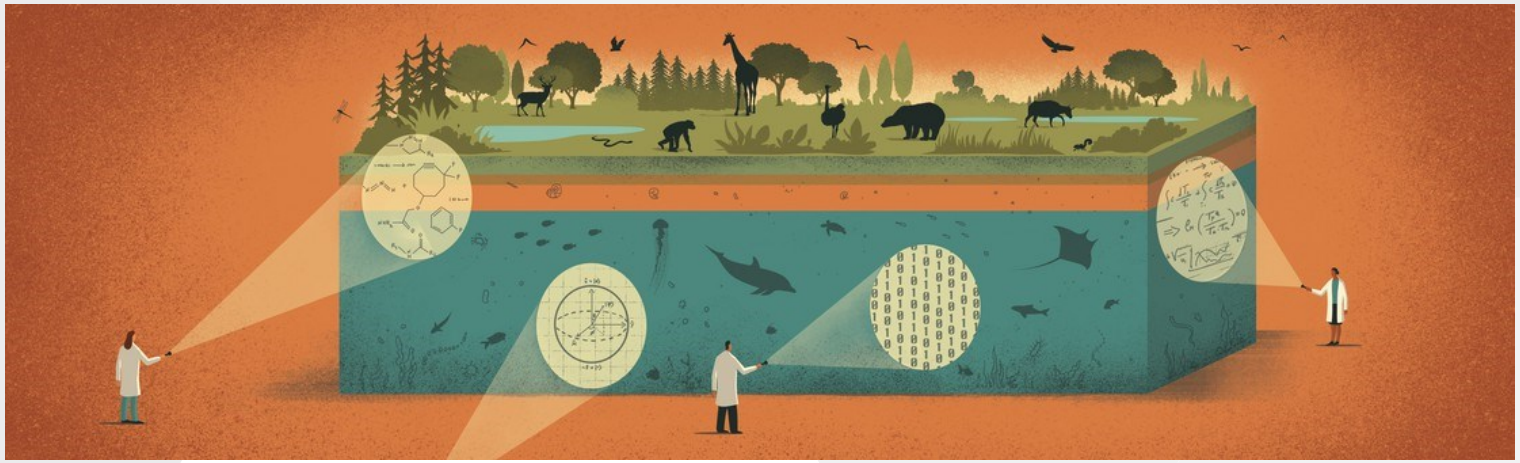
- Spoiler alert! Not many equations today
  - Understanding key things, not all related equations

- Virtually no student in history has understood every equation in a course without doing homework, whether assigned or self-initiated
  - Literature:
    - Frenkel & Smit “Understanding Molecular Simulation. From Algorithms to Applications”
    - Alen & Tildesley “Computer Simulations of Liquids”
    - ChatGTP

- Prioritize critical thinking and informed learning

Why Physics?

- Student study programs:
  - Life Sciences
  - Biochemistry
  - Bioinformatics
  - Environmental Health Sciences
  - International Relations and European Politics



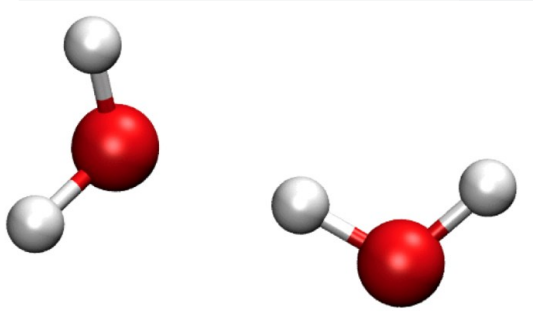
from https://elifesciences.org/subjects/physics-living-systems

- No single mention of physics. Why do we need it?

- Physics is always behind:
  - Structural organization in biological systems (cell membranes, extracellular matrix, etc.)
  - Interactions of biomolecules (proteins, drugs, etc.)
  - Even European politician obey the laws of physics (unlike many others)

Quantum physics?

- Biosimulations usually do not have electrons. Is it legal?
  - Atoms are represented as billiard balls
  - Averaged effects of electrons are considered through a set of parameters (force fields)
  - Force fields define the potential energy of a system as a function of atomic positions



- This approximation does not work:
  - Electron transfer processes
  - Reactive systems
  - Strong quantum effects

- Use instead: ab initio molecular dynamics (AIMD), quantum mechanics/molecular mechanics (QM/MM) methods

Born-Oppenheimer approximation





Born-Oppenheimer approximation





Born-Oppenheimer approximation

Born-Oppenheimer approximation:

- $M_e << M_{nucl}$ ⇒ nuclei move in an averaged field of electrons ⇒ motion of electrons can be separated from motion of nuclei

$$\Psi_{tot} = \psi_{el}(r;R)\psi_n(R)$$

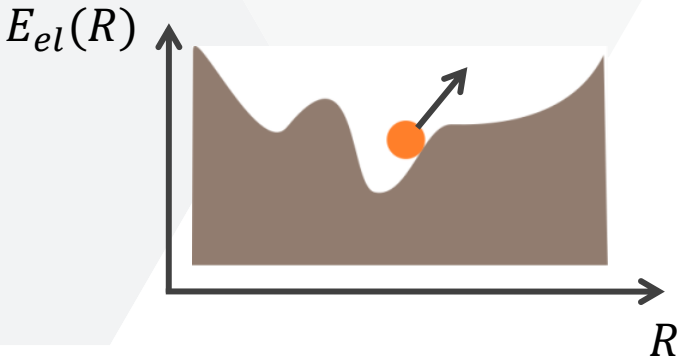
for electrons:

$$H_{el}(r;R)\psi_{el}(r;R) = E_{el}(R)\,\psi_{el}(r;R)$$

for nuclei:

$$[T_n + E_{el}(R)]\psi_n(R) = \mathrm{E}\psi_n(R)$$

- Nuclei move on a potential energy surface $E_{el}(R)$, which can be also denoted as $V(\vec{r})$
- System remains in one electronic state
- Fails: degenerate energy levels



- Classical mechanics can be used for nuclei
  - de Broglie wavelength (represents a wave-like property of particles) is exceedingly small compared to the size of nuclei
- Not the case of an electron

Born-Oppenheimer approximation

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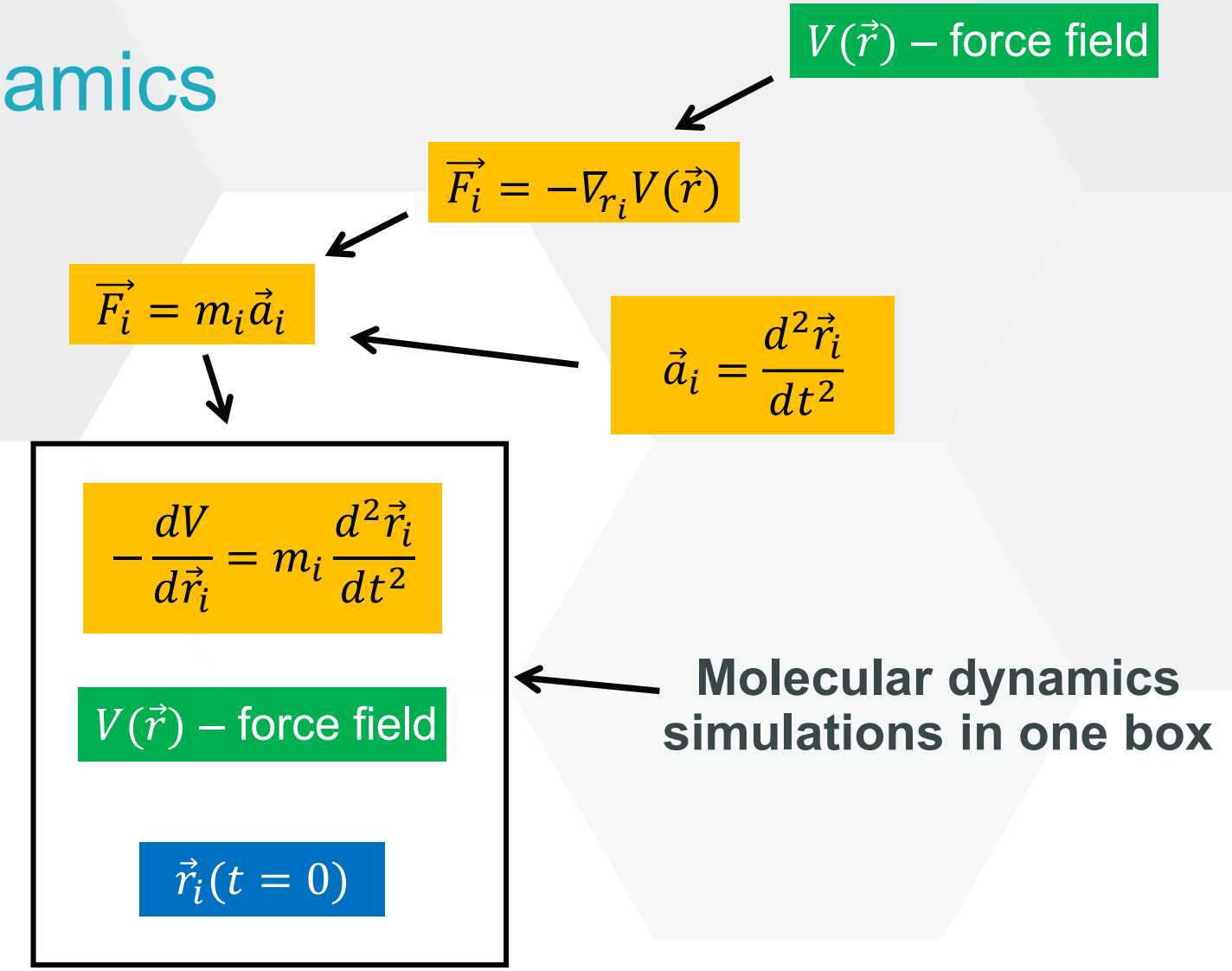
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- $V(\vec{r})$ – force field (a set of predefined parameters) in classical molecular dynamics
  - **Used in biosimulations!**
- … or can be calculated on-the-fly from the electronic wave function in *ab initio* molecular dynamics simulations
  - Not used in biosimulations, only small systems possible, computationally very expensive, short simulation times, …

Classical molecular dynamics

- Classical Newton's equations of motion
- Trajectory (position of atoms in time) is the output of a simulation.
- Motion of atoms is dictated by force field (interaction parameters)
- Initial conditions must be set – initial position of atoms, for instance, protein structure from Protein Data Bank or AlphaFold



Classical molecular dynamics

The force field (V) and the environment (r) determine the force (F) that an atom feels

The force (F) experienced by an atom and its state determine how an atom moves (a)

$V(\vec{r})$ – force field

$$\overrightarrow{F_i} = -\nabla_{r_i} V(\vec{r})$$

$$\overrightarrow{F_i} = m_i \vec{a}_i = m_i \frac{d^2 \vec{r}_i}{dt^2}$$

Force fields: not today :)

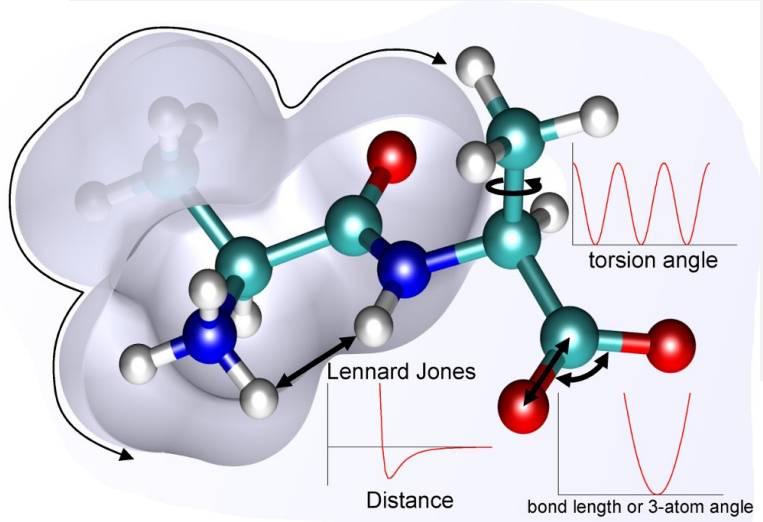
- Week 4:
  - General principles of a force field
  - Nonbonded interactions: van der Waals and electrostatics
  - Cut-offs, Ewald summation, periodic boundary conditions



- Week 6:
  - Force fields in biosimulations
  - Partial charges
  - Developing a new force field for a drug



Ondřej Kroutil



Solving equations of motion

$$m_i \frac{d^2 \vec{r}_i}{dt^2} = \overrightarrow{F_i} = -\nabla_{r_i} V(\vec{r})$$

Solving equations of motion

- Analytical solutions are unrealistic, too many particles

$$m_i \frac{d^2\vec{r}_i}{dt^2} = \overrightarrow{F_i} = -\nabla_{r_i} V(\vec{r})$$

- Numerical integration of Newton's equations

- Things to be considered:
  - accuracy
  - time irreversibility
  - numerical stability
  - computational efficiency

- Timescale of biosimulations: ~1 μs = 10^{-6} s
- Typical time step (next slides): 2 fs = 10^{-15} s

- Simulations requires **500,000,000** time steps
  - calculated for hundreds of thousands of particles in system

Taylor’s expansion of equations of motion

$$\dot{r}_i(t) \qquad \frac{f_i(t)}{m} = a_i(t) = \ddot{r}_i(t)$$

$$\mathbf{r}_i(t+\Delta t) = \mathbf{r}_i(t) + \Delta t\ \mathbf{v}_i(t) + \frac{\Delta t^2}{2m_i}\ \mathbf{f}_i(t) + \frac{\Delta t^3}{3!}\ \dddot{\mathbf{r}}_i(t) + \mathcal{O}(\Delta t^4)$$

$$\mathbf{v}_i(t+\Delta t) = \mathbf{v}_i(t) + \frac{\Delta t}{m_i}\ \mathbf{f}_i(t) + \frac{\Delta t^2}{2}\ \ddot{\mathbf{v}}_i(t) + \frac{\Delta t^3}{3!}\ \dddot{\mathbf{v}}_i(t) + \mathcal{O}(\Delta t^4)$$

Euler algorithm

- Linear approximation
  - forward difference approximation

$$\mathbf{r}_i(t+\Delta t) = \mathbf{r}_i(t) + \Delta t\ \mathbf{v}_i(t) + \frac{\Delta t^2}{2m_i}\mathbf{f}_i(t) + \mathcal{O}(\Delta t^3)$$

$$\mathbf{v}_i(t+\Delta t) = \mathbf{v}_i(t) + \frac{\Delta t}{m_i}\ \mathbf{f}_i(t) + \mathcal{O}(\Delta t^2)$$

- Not used in molecular dynamics
  - accumulation of errors over time
  - time irreversible (Newton's equations are)

Euler algorithm

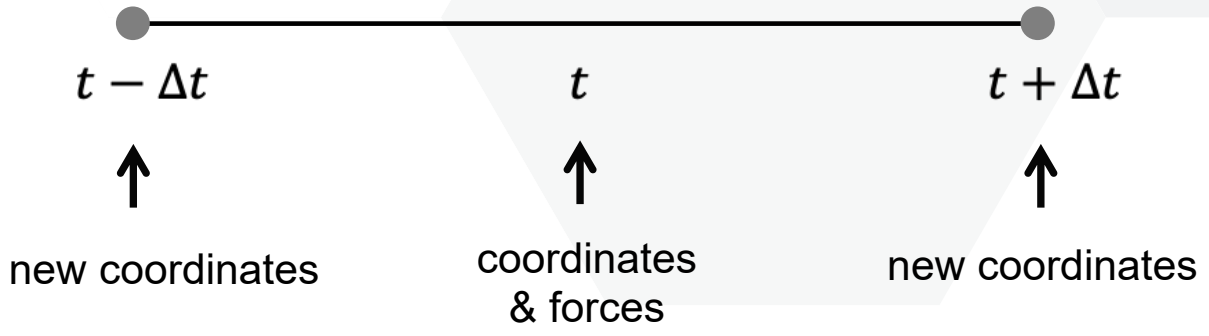
- Linear approximation
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$$\mathbf{r}_i(t+\Delta t) = \mathbf{r}_i(t) + \Delta t\ \mathbf{v}_i(t) + \frac{\Delta t^2}{2m_i}\mathbf{f}_i(t) + \mathcal{O}(\Delta t^3)$$

$$\mathbf{v}_i(t+\Delta t) = \mathbf{v}_i(t) + \frac{\Delta t}{m_i}\ \mathbf{f}_i(t) + \mathcal{O}(\Delta t^2)$$

- Not used in molecular dynamics
  - accumulation of errors over time
  - time irreversible (Newton's equations are)

- Trick to introduce time reversibility and improve approximation:
  - Two points in time used to predict the next one



Verlet algorithm

$$\mathbf{r}_i(t+\Delta t) = \mathbf{r}_i(t) + \Delta t\ \mathbf{v}_i(t) + \frac{\Delta t^2}{2m_i}\ \mathbf{f}_i(t) + \frac{\Delta t^3}{3!}\ \dddot{\mathbf{r}}_i(t) + \mathcal{O}(\Delta t^4)$$

$$\mathbf{r}_i(t-\Delta t) = \mathbf{r}_i(t) - \Delta t\ \mathbf{v}_i(t) + \frac{\Delta t^2}{2m_i}\ \mathbf{f}_i(t) - \frac{\Delta t^3}{3!}\ \dddot{\mathbf{r}}_i(t) + \mathcal{O}(\Delta t^4)$$

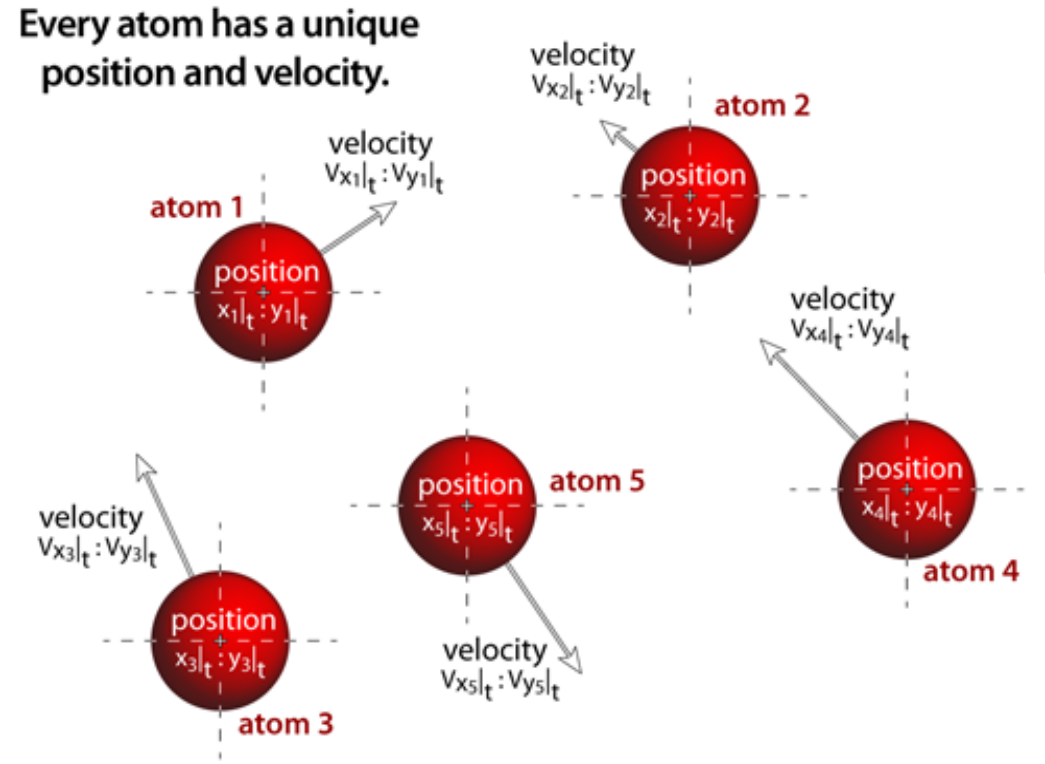
(i) positions from adding and (ii) velocities from subtracting the above equations:



$$\mathbf{r}_i(t+\Delta t) = 2\mathbf{r}_i(t) - \mathbf{r}_i(t-\Delta t) + \frac{\Delta t^2}{m_i}\mathbf{f}_i(t) + \mathcal{O}(\Delta t^4)$$

$$\mathbf{v}_i(t) = \frac{\mathbf{r}_i(t+\Delta t) - \mathbf{r}_i(t-\Delta t)}{2\Delta t} + \mathcal{O}(\Delta t^3).$$

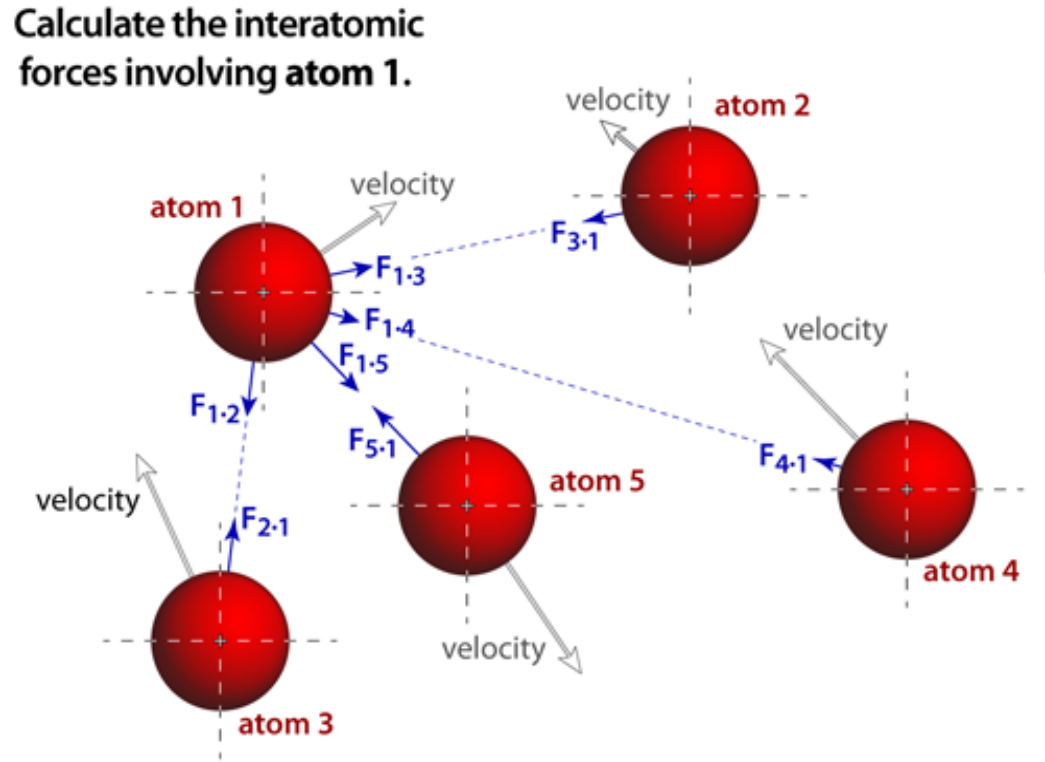
Time Evolution: Step 1



- Every atom has a unique position (in this example, x and y) and velocity (v_x and v_y)
- They move according to Newton's laws of motion
- Acceleration ($\vec{a}$) acts on every atom and is determined by the force ($\vec{F}$) acting on the atom divided by the atom's mass (m).

http://atomsinmotion.com/book/chapter5/md

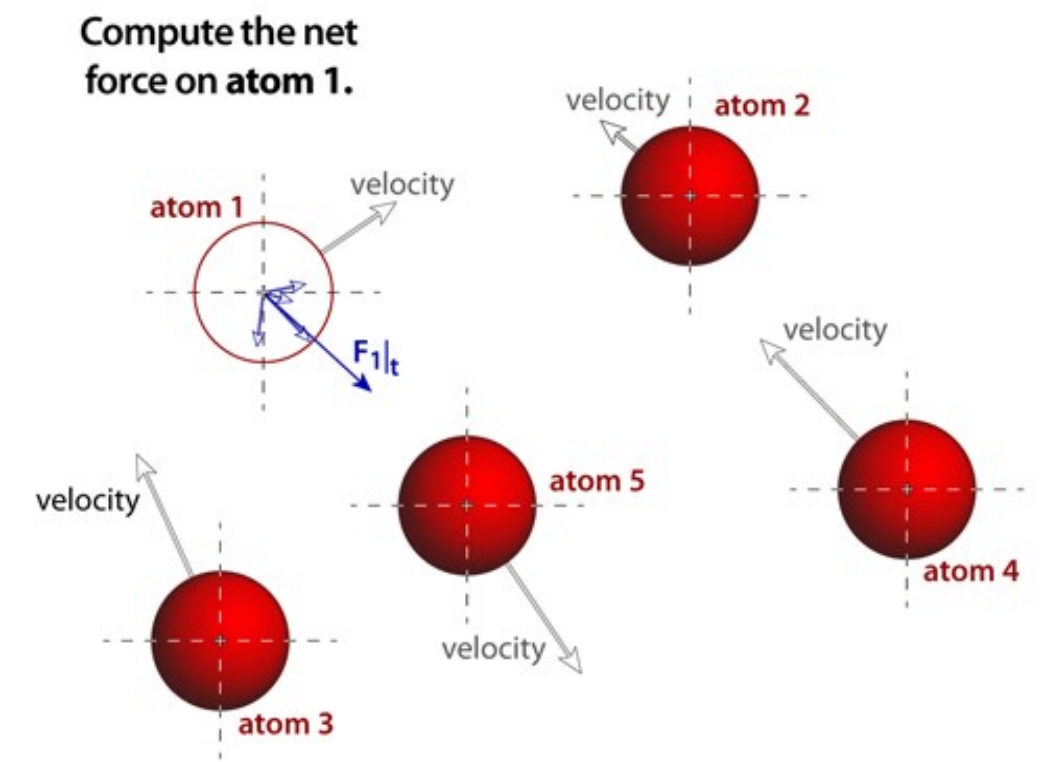
Time Evolution: Step 2



- Force is determined by the force field and contributions from every atom are considered
- Newton's third law of motion holds too

http://atomsinmotion.com/book/chapter5/md

Time Evolution: Step 3



- Force ($\vec{F}$) acting on any atom is the vector sum of the individual forces that arise from the atomic pairs
- Defined by the force field:
  - Intramolecular interactions (bonds, angles, etc.)
  - Intermolecular interactions (van der Waals, electrostatics)
  - Weeks 4 and 6

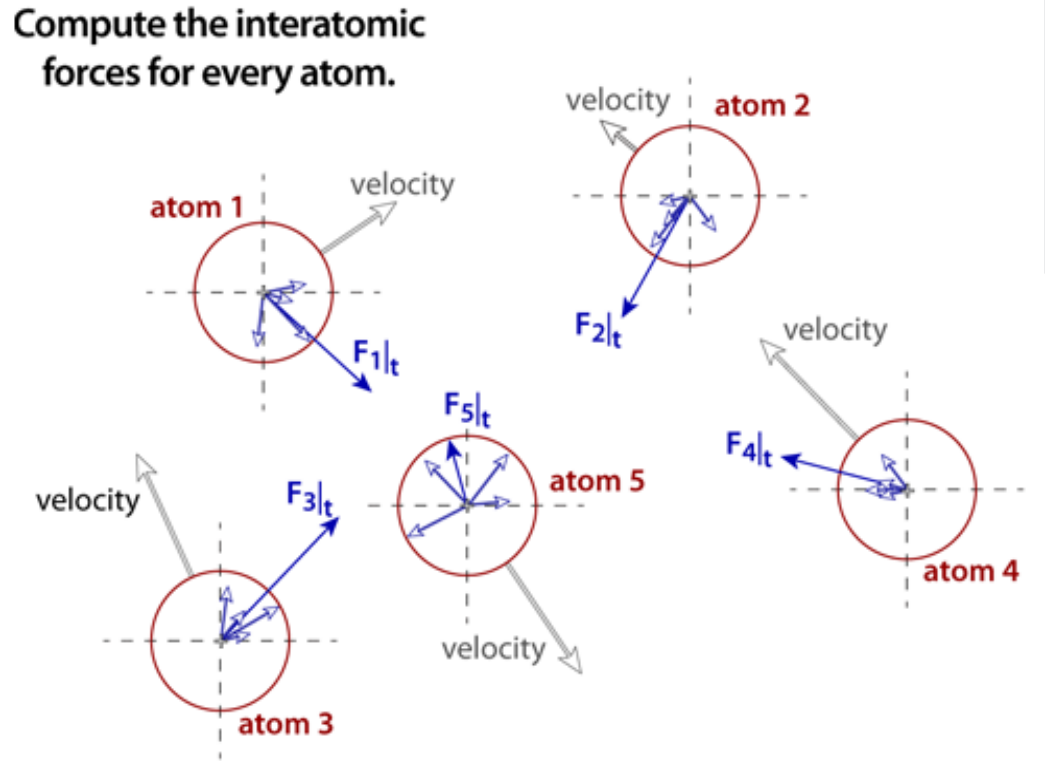
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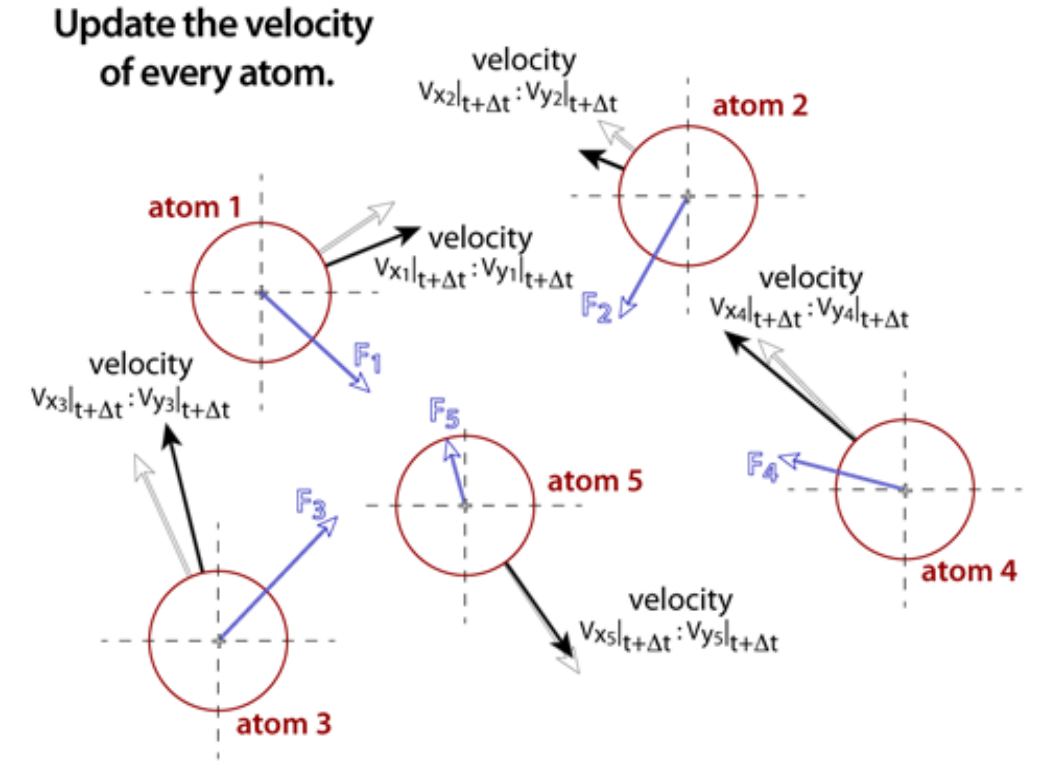
Time Evolution: Step 4



- Force ($\vec{F}$) is calculated for every atom
- All pairwise contributions are taken into account:
  - Explains the amount of calculations that must be done

http://atomsinmotion.com/book/chapter5/md

Time Evolution: Step 5



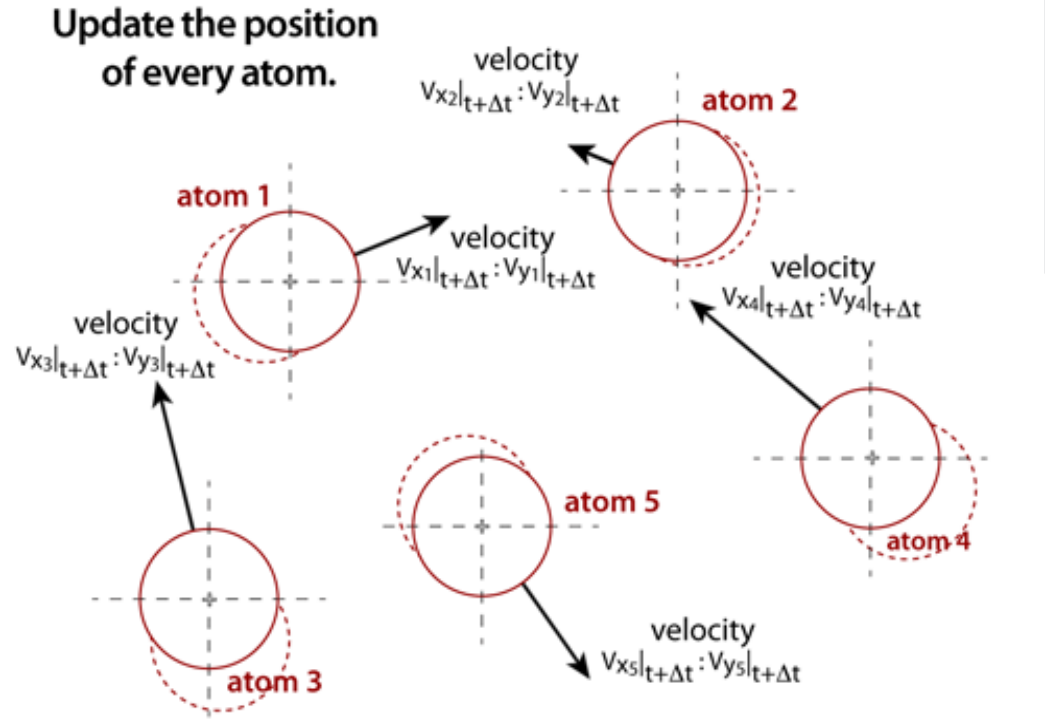
http://atomsinmotion.com/book/chapter5/md

- Solving equations of motions (previous slides)
- Velocities are calculated for every atom
- Velocities are further can be changed due to thermostat to maintain the appropriate energy of the system:
  - Week 5



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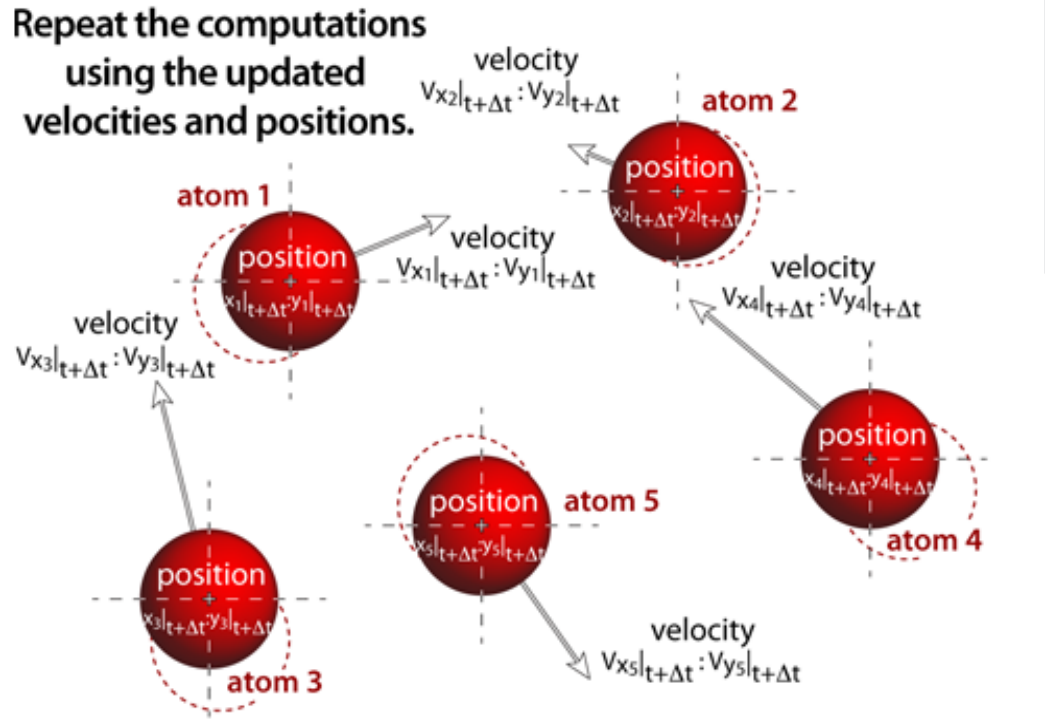
Time Evolution: Step 6



- Positions are updated
- All the information is in play: forces, velocities, time step

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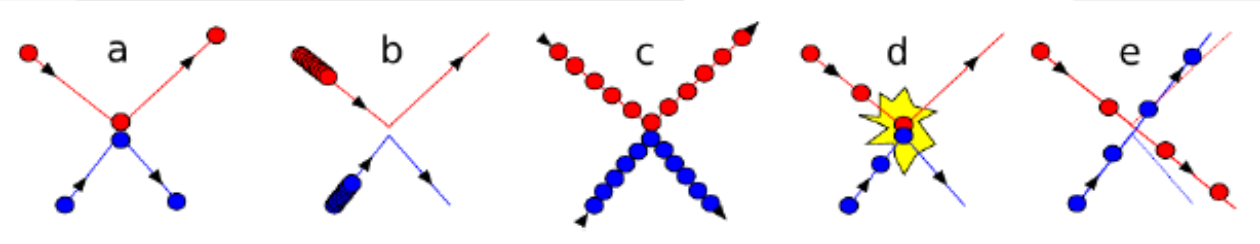
Time Evolution: Step 7



- The concept of molecular dynamics simulations: output from one iteration is an input for the next one:
  - return to step 1
- Keep going until when? Total simulations time?
- What about time step?

http://atomsinmotion.com/book/chapter5/md

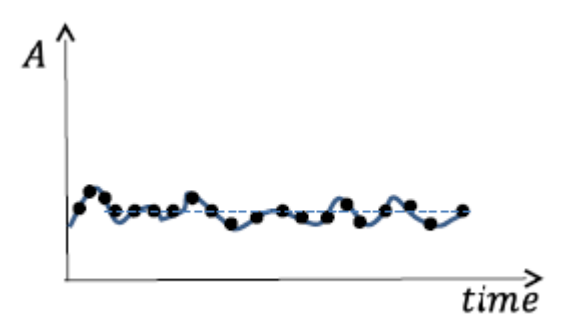
Time step: optimal Δt



- Time step: not too big, not too small
- Depends on the force field and its implementation
- Rule of thumb: at least, one order of magnitude shorter than a characteristic time of the fastest motion in the system
  - vibration of bonds, time scale ~10^{-14} s, *i.e.*, time step ~1–5 fs

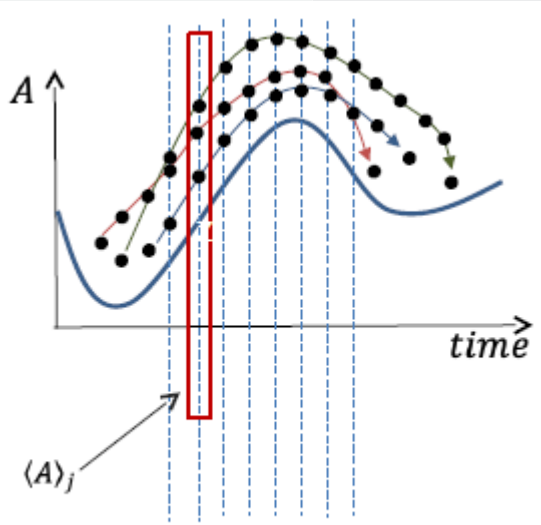
Time in simulations

- How long simulation should be?
  - Systems under equilibrium and out of equilibrium
- Many processes are meaningful only over certain period of time
  - Chemical reactions (~10^{-15} s – seconds)
  - Amino acid and protein conformation (~10^{-12} s – 10^{-6} s)
  - Protein adsorption to the membrane / protein conformation within the membrane (~10^{-6} s – 10^{-3} s)
- Current must-have in biological simulations – order of microseconds (10^{-6} s)



$$\langle A \rangle_{time} = \frac{1}{N} \sum_{i=1}^{N} A(t_i)$$

under equilibrium



$$\langle A(t) \rangle_{set} = \langle A \rangle_j (t_i)$$

out of equilibrium

Time in simulations

- Under equilibrium
  - One simulation/trajectory is enough (should be long enough though)
  - Ergodicity (all accessible states are equally visited)
  - Average over ensemble = average over time (next week to discuss)
  - System properties are calculated as averages in time
  - Time is not important *per se*

- Out of equilibrium
  - Many simulations/trajectories are needed
  - One trajectory – one possible path
  - System properties are calculated as averages over a set of trajectories
  - Time is important

Real example and today practical session

from https://www.easybiologyclass.com



a molecular simulation of ink dissolving in water
is a system out of equilibrium

a molecular simulation of this liquid
is a system under equilibrium

Break!

Get ready for practical session!

Get ready for practical session

- Wolf computers with GPU (more comfortable to work on)
  - wolf04
  - wolf05
  - wolf07
  - wolf08
  - wolf09
  - wolf12
  - wolf18
  - wolf19

Get ready for practical session

- A bit of things to be done:
  - `cd`
  - `mkdir C9555/`
  - `mv ~/survival_guide_simulations_c9555/week01/ ~/C9555/`
  - `rm -rf {SURVIVAL_GUIDE_DIRECTORY}`
    - **BE SURE THAT PREVIOUS COMMAND (MV) WAS SUCCESSFUL BEFORE DELETING**

- https://gitlab.com/denysbiriukov/survival_guide_simulations_c9555

Get ready for practical session

- `git clone https://gitlab.com/denysbiriukov/survival_guide_simulations_c9555.git`
- `cd survival_guide_simulations_c9555/`
- `cd week02/`
- `bash ../infinity/infinity-fix-autoload.sh` (optional)
- `bash ../infinity/infinity-temp.sh`
- `bash ../conda/conda-adjust.sh`
  - if it does not work, first do
    `bash ../conda/conda-light-activate.sh`
- `bash ../conda/jupyter-start.sh`